

Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) with hands-on experience automating data quality and reporting processes for regulated financial data at Deutsche Bundesbank, using Python and SQL to replace manual, case-by-case checks with parameterized, reusable logic. Complements this with independent research in lattice-based cryptography, bringing quantitative rigor to problems in finance and security. Comfortable working close to financial markets data end-to-end: validation, automation, and clear documentation for stakeholders.

Core Competencies

- **Programming:** Python, SQL, R, Java, C++
- **Financial & Regulatory Data:** Automated validation and error detection on supervisory reporting data at Deutsche Bundesbank; parameterized, reusable query logic applicable across datasets and reporting periods
- **Process Automation:** Converted case-by-case manual scripts into configuration-driven, batch-capable Python/SQL solutions; automated ticketing and stakeholder notification workflows
- **Quantitative Analysis:** Complexity and security-parameter analysis in lattice-based cryptography research; statistical benchmarking (percentile-based anomaly detection) over rolling time windows
- **Databases & Tools:** SQL, SQLite, MySQL, Excel, Git
- **Languages:** English (C1, TOEFL iBT 105/120), German (C1), Spanish (native)

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern *04/2026–07/2026*
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to ensure identical business logic after data migration
 - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations), with automated Jira ticketing and emails to reporting banks
 - Worked within an agile Scrum framework (sprint planning, daily stand-ups, sprint reviews)

Automated Validation Framework for Regulatory Reporting Data – converted a non-scalable, case-by-case validation script into a parameterized, reusable Python solution

 - Designed parameterized MySQL queries applicable to any database, attribute, and reporting period, replacing hard-coded per-case script logic
 - Extended the validation logic with percentile-based statistical benchmarking over a rolling 12-month window to flag anomalous reporting frequencies and separate false positives from genuine data-quality issues
 - Built a YAML-driven configuration interface enabling batch validation across multiple databases, attributes, and reporting periods in a single call

Projects

- **Personal Project (GitHub)**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Built an end-to-end pipeline for simulated bank transactions (150 accounts) with a normalized SQLite database, foreign-key integrity checks, and parameterized queries throughout
 - Staged data cleaning with a fully auditable log; automated balance maintenance via SQL triggers
- **Tecnológico de Monterrey**
Lattice-Based Cryptography Research 10/2024–02/2025
 - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python; contributed to a scientific publication
- **Tecnológico de Monterrey**
Predictive Modeling – Healthcare Optimization 10/2024–12/2024
 - Analyzed 25+ years of Mexican national health data to identify patient risk profiles and resource bottlenecks; built predictive models (Random Forest, Neural Networks) and K-Means clustering for high-risk segmentation

Education

- **Tecnológico de Monterrey** Mexico
BSc in Data Science 2022–2027
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, cryptography & security, numerical optimization.
- **Universität Göttingen** Germany
Exchange Semester 2025–2026

Languages

- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.